Discrete I — Logic, Sets & Proofs

algebra refreshers

Primary-school tools you'll need in every proof. Referenced throughout this section.

Exponent rules

- $x^0=1 \quad (x \neq 0), \quad x^1=x$
- $x^{-n}=\frac{1}{x^n}, \quad \left(\frac{a}{b}\right)^{-n}=\left(\frac{b}{a}\right)^n$
- $x^{m+n}=x^m \cdot x^n, \quad x^{m-n}=\frac{x^m}{x^n}$
- $(x^m)^n=x^{mn}, \quad (xy)^n=x^n y^n, \quad (x/y)^n=x^n/y^n$
- $x^{m/n}=\sqrt[n]{x^m}=(\sqrt[n]{x})^m$
- $b^{k+1}=b \cdot b^k$ (useful for induction)
- $(-1)^{\text{even}}=1, \quad (-1)^{\text{odd}}=-1, \quad (-1)^{k+1}=-(-1)^k$

Polynomial multiplication

- **FOIL** $(a+b)(c+d)=ac+ad+bc+bd$
- $(a+b)^2=a^2+2ab+b^2$
- $(a-b)^2=a^2-2ab+b^2$
- $(a+b)(a-b)=a^2-b^2$ (diff. of squares)
- $(a+b)^3=a^3+3a^2b+3ab^2+b^3$
- $(a-b)^3=a^3-3a^2b+3ab^2-b^3$
- $(a+b+c)^2=a^2+b^2+c^2+2(ab+ac+bc)$

Distribute to *every* term: $a(b+c+d)$ has 3 outputs, not 2.

Factoring patterns

- **GCF** first: $6x^3+9x^2=3x^2(2x+3)$
- **Diff. of squares**: $a^2-b^2=(a+b)(a-b)$
- **Perfect sq. trinomial**: $a^2 \pm 2ab+b^2=(a \pm b)^2$
- **Sum of cubes**: $a^3+b^3=(a+b)(a^2-ab+b^2)$
- **Diff. of cubes**: $a^3-b^3=(a-b)(a^2+ab+b^2)$
- **Quadratic** x^2+bx+c: find p,q with $p+q=b$, $pq=c \Rightarrow (x+p)(x+q)$
- **Grouping**: $ax+ay+bx+by=a(x+y)+b(x+y)=(a+b)(x+y)$

Which method? — decide by counting terms

1. **GCF first, always.** Pull out the biggest factor every term shares. Check: multiply back to confirm.
2. **Count the terms:**
   - 2 terms → diff. of squares a^2-b^2, or sum/diff. of cubes.
   - 3 terms → trinomial. Perfect square? Else do the p,q search below.
   - 4 terms → grouping.

Worked: quadratic x^2+5x+6

Goal: two numbers p,q that multiply to c and add to b.

1. LIST FACTOR PAIRS OF $c=6$
$1 \cdot 6, \quad 2 \cdot 3$

↓

2. WHICH PAIR ADDS TO $b=5$?
$2+3=5$ ✓$\Rightarrow p=2, q=3$

↓

3. WRITE IT
$(x+2)(x+3)$

↓

4. FOIL BACK TO VERIFY
$x^2+3x+2x+6=x^2+5x+6$ ✓

Sign rules (read off c then b): $c>0$: p,q same sign (matching b); $c<0$: opposite signs, the bigger takes b's sign. **Worked,** $c<0$: x^2-x-12: pairs of -12 summing to -1: $-4,+3 \Rightarrow (x-4)(x+3)$.

Worked: $a \neq 1$ *— the AC method on* $6x^2+11x+3$

1. MULTIPLY $a \cdot c$
$6 \cdot 3=18$; need a pair with product 18, sum $b=11$: 2 and 9

↓

2. SPLIT THE MIDDLE TERM
$6x^2+2x+9x+3$

↓

3. GROUP
pair off: $(6x^2+2x)+(9x+3)$; GCF each pair: $2x(3x+1)+3(3x+1)$

↓

4. FACTOR THE SHARED BINOMIAL
$(2x+3)(3x+1)$; FOIL back ✓ *(sitting next to the $a=1$ walkthrough: same p,q hunt, against ac instead of c)*

Worked examples, the remaining patterns

- **GCF**: $8x^3-12x^2=4x^2(2x-3)$. (shared: $4x^2$)
- **Diff. squares**: $9x^2-25=(3x)^2-(5)^2=(3x+5)(3x-5)$.
- **Perfect sq. trinomial**: $x^2-10x+25$: middle $=2 \cdot 5 \cdot x$ ✓$\Rightarrow (x-5)^2$.
- **Sum of cubes**: $8x^3+27=(2x)^3+3^3=(2x+3)(4x^2-6x+9)$.
- **Diff. of cubes**: $x^3-64=x^3-4^3=(x-4)(x^2+4x+16)$. *(SOAP: Same sign, Opposite, Always Plus)*

Worked: grouping — x^3+2x^2+3x+6

1. SPLIT INTO PAIRS
$(x^3+2x^2)+(3x+6)$

↓

2. GCF EACH PAIR
$x^2(x+2)+3(x+2)$ — the binomials must match, or regroup

↓

3. FACTOR THE SHARED BINOMIAL
$(x^2+3)(x+2)$

Common mistakes: dropping a sign when distributing the minus. Only canceling *factors*, never *terms* (see **Simplifying fractions** below); forgetting to pull the GCF before pattern-matching.

Fraction operations

- $\frac{a}{b}+\frac{c}{d}=\frac{ad+bc}{bd}$ (common denom.)
- $\frac{a}{b} \cdot \frac{c}{d}=\frac{ac}{bd}$
- $\frac{a}{b} \div \frac{c}{d}=\frac{a}{b} \cdot \frac{d}{c}=\frac{ad}{bc}$
- Non-fraction + fraction: $\frac{A}{B}+C=\frac{A+CB}{B}$
- Complex fraction: $\frac{a/b}{c/d}=\frac{a}{b} \cdot \frac{d}{c}=\frac{ad}{bc}$

Simplifying fractions: factors against terms

Only cancel factors, never terms. A factor multiplies the whole numerator or denominator. A term is added/subtracted.

- **OK**: $\frac{x(x+1)}{x(x+2)}=\frac{x+1}{x+2}$ (cancel factor x)
- **NOT OK**: $\frac{x+1}{x+2} \neq \frac{1}{2}$ (can't cancel term x)

Always factor numerator and denominator fully before canceling.

Common denominators for induction

Same base, different exponents: pick the larger exponent.

$$\frac{A}{b^k}+\frac{B}{b^{k+1}}=\frac{Ab+B}{b^{k+1}}$$

Shared factor in denominator: pick the one with more factors.

$$\frac{A}{(k+1)}+\frac{B}{(k+1)(k+2)}=\frac{A(k+2)+B}{(k+1)(k+2)}$$

The concrete case induction actually needs (telescoping-sum inductive step, worked below):

$$\frac{k}{k+1}+\frac{1}{(k+1)(k+2)}=\frac{k(k+2)+1}{(k+1)(k+2)}=\frac{k^2+2k+1}{(k+1)(k+2)}$$

$$=\frac{(k+1)^2}{(k+1)(k+2)}=\frac{k+1}{k+2}$$

The expanded top must factor with a $(k+1)$ — if it doesn't, the algebra slipped.

Solving equations

- **Linear** $ax+b=c$: isolate x.
- **Quadratic formula** $ax^2+bx+c=0$: $x=\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$
- **Discriminant** $\Delta=b^2-4ac$: >0 two real roots; $=0$ one; <0 two complex.
- **Completing the square**: $x^2+bx=(x+\frac{b}{2})^2-(\frac{b}{2})^2$
- **Radical**: isolate root, square both sides, check for extraneous solutions.

One quadratic, both routes — $x^2-6x+2=0$. Formula:

$$x=\frac{6\pm\sqrt{36-8}}{2}=\frac{6\pm\sqrt{28}}{2}=\frac{6\pm2\sqrt{7}}{2}=3\pm\sqrt{7}$$

Completing the square:
$x^2-6x=-2 \Rightarrow (x-3)^2-9=-2 \Rightarrow (x-3)^2=7 \Rightarrow x=3\pm\sqrt{7}$ — same roots, and the second route also hands over the vertex.

Worked: radical equation — the extraneous root, live

$\sqrt{x+7}=x-5$:

1. SQUARE BOTH SIDES
$x+7=x^2-10x+25$

↓

2. SOLVE THE QUADRATIC
$x^2-11x+18=0=(x-9)(x-2) \Rightarrow x=9,\ 2$

↓

3. CHECK BOTH — MANDATORY
$x=9$: $\sqrt{16}=4=9-5$ ✓; $x=2$: $\sqrt{9}=3$ vs $2-5=-3$ fails ⇒ extraneous. Answer: $x=9$ only *(squaring created the ghost — the check is part of the solve)*

Absolute value

$$|x|=\begin{cases}x & x\geq 0\\ -x & x<0\end{cases}$$

- $|ab|=|a||b|$, $|a/b|=|a|/|b|$
- $|a+b|\leq|a|+|b|$ (triangle inequality)
- $|x|<c \iff -c<x<c$
- $|x|>c \iff x>c$ or $x<-c$
- $|x|=c \iff x=\pm c$

The </> pair, worked: $|2x-3|<5$: $-5<2x-3<5$; add 3 across: $-2<2x<8$; divide by 2: $-1<x<4$ (one band). $|2x-3|>5$: $2x-3>5 \Rightarrow 2x>8 \Rightarrow x>4$, *or* $2x-3<-5 \Rightarrow 2x<-2 \Rightarrow x<-1$: $x>4$ or $x<-1$ (two tails). Less-than = between. Greater-than = or — never chain the $>$ case into one inequality.

Inequalities

- Add/subtract same value: preserves direction.
- Multiply/divide by positive: preserves direction.
- **Multiply/divide by negative: FLIP direction.**
- Transitivity: $a<b$ and $b<c \Rightarrow a<c$.
- Compound $a<x<b$: treat as two inequalities simultaneously.
- Squaring: preserves if both sides nonneg. Otherwise dangerous.

The flip, live: $-2x<6$: divide by -2: $x>-3$. Sanity: $x=0$ satisfies both ✓— forgetting the flip gives $x<-3$, which 0 instantly refutes: test one number, always.

Logarithms

$\log_b(x)=y \iff b^y=x$. Bases: $\log=\log_{10}$, $\ln=\log_e$.

- $\log_b(xy)=\log_b x+\log_b y$
- $\log_b(x/y)=\log_b x-\log_b y$
- $\log_b(x^n)=n\log_b x$
- $\log_b(1)=0$, $\log_b(b)=1$
- **Change of base**: $\log_b x=\dfrac{\log_c x}{\log_c b}$
- $b^{\log_b x}=x$, $\log_b(b^x)=x$

Example: $\log_2 9=\frac{\ln 9}{\ln 2}\approx\frac{2.197}{0.693}\approx 3.17$ — sanity: $2^3=8<9<16=2^4$, so between 3 and 4 ✓.

Sigma notation

$\sum_{i=m}^{n} a_i=a_m+a_{m+1}+\cdots+a_n$. Gives **partial sum**, not individual terms.

- **Constant**: $\sum_{i=1}^{n} c=cn$
- **Pull constant**: $\sum c\cdot a_i=c\sum a_i$
- **Split**: $\sum(a_i\pm b_i)=\sum a_i\pm\sum b_i$
- **Shift index**: $\sum_{i=m}^{n} a_i=\sum_{j=0}^{n-m} a_{j+m}$
- **Separate last term**: $\sum_{i=1}^{n+1} a_i=\sum_{i=1}^{n} a_i+a_{n+1}$

Classic sums:

- $\sum_{i=1}^{n} i=\dfrac{n(n+1)}{2}$
- $\sum_{i=1}^{n} i^2=\dfrac{n(n+1)(2n+1)}{6}$
- $\sum_{i=1}^{n} i^3=\left(\dfrac{n(n+1)}{2}\right)^2$
- $\sum_{i=1}^{n}(2i-1)=n^2$ (odds)
- $\sum_{i=0}^{n} r^i=\dfrac{r^{n+1}-1}{r-1}$ ($r\neq 1$, geometric)
- $\sum_{i=1}^{n}\dfrac{1}{i(i+1)}=\dfrac{n}{n+1}$ (telescoping)

Manipulation worked:
$\sum_{i=1}^{n}(2i+3)=2\sum i+\sum 3=2\cdot\frac{n(n+1)}{2}+3n=n(n+1)+3n=n^2+n+3n=n^2+4n$ (check $n=2$: $5+7=12=4+8$ ✓). **Index shift**: $\sum_{i=3}^{7} i^2=\sum_{j=0}^{4}(j+3)^2$ — substitute $j=i-3$ and shift both bounds and the body.

propositional logic

Propositions & connectives

Statement that is either true (T) or false (F), never both. Variables: $p,q,r,\ldots$

- **negation** $\neg p$: "not p"; true when p false.
- **conjunction** $p\wedge q$: "and" (but, however, yet); true iff both true.
- **disjunction** $p\vee q$: "or" (inclusive); true iff at least one true.
- **conditional** $p\to q$: "if p then q"; false only when p true and q false.
- **biconditional** $p\leftrightarrow q$: "iff"; true iff same truth value.
- **xor** $p\oplus q$: true iff exactly one true.

Translation example: "If it rains but I have an umbrella, I stay dry." $p=$ rain, $q=$ umbrella, $r=$ dry: $(p\wedge q)\to r$ — "but" is an $\wedge$ in disguise.

Truth table

p	q	$p\wedge q$	$p\vee q$	$p\to q$	$p\leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Worked: build a compound table — $(p \wedge q) \to \neg r$

1. COUNT ROWS
3 variables $\Rightarrow 2^3=8$ rows. Alternate r fastest, p slowest

↓

2. ONE COLUMN PER SUBFORMULA
build $p \wedge q$, then $\neg r$, then the arrow — never the whole thing in one pass. The decisive rows:

p	q	r	$p \wedge q$	$\neg r$	$(p \wedge q) \to \neg r$
T	T	T	T	F	F
T	T	F	T	T	T
F	.	.	F	.	T

(any row with $p \wedge q=F$ is vacuously T — the arrow only breaks on a kept promise)

↓

3. FILL & CLASSIFY
F needs $p \wedge q=T$ (forces $p=q=T$) *and* $\neg r=F$ (forces $r=T$): the single row TTT. Row TTF gives T: both values occur $\Rightarrow$ contingency (not a tautology, not a contradiction)

Conditional variants

Given $p \to q$:

- **converse**: $q \to p$
- **inverse**: $\neg p \to \neg q$
- **contrapositive**: $\neg q \to \neg p$ $\quad (\equiv p \to q)$

Worked — one sentence, four forms: "If n is divisible by 4, then n is even" (true):

- converse: "if even, then divisible by 4" — false ($n=6$)
- inverse: "if not divisible by 4, then odd" — false ($n=6$ again: converse and inverse are each other's contrapositives)
- contrapositive: "if odd, then not divisible by 4" — true, always matches the original

Necessary vs. sufficient

- p **sufficient** for q: $p \to q$
- p **necessary** for q: $q \to p$
- p iff q: necessary *and* sufficient

Example: divisible-by-4 is sufficient for even (it guarantees it) but not necessary (6 is even without it). Even is necessary for divisible-by-4.

Logical equivalences

- **Tautology** (always T): $p \vee \neg p$
- **Contradiction** (always F): $p \wedge \neg p$
- **Identity**: $p \wedge T \equiv p$, $\quad p \vee F \equiv p$
- **Domination**: $p \vee T \equiv T$, $\quad p \wedge F \equiv F$
- **Idempotent**: $p \wedge p \equiv p$, $\quad p \vee p \equiv p$
- **Double neg.**: $\neg(\neg p) \equiv p$
- **Commutative**: $p \wedge q \equiv q \wedge p$
- **Associative**: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- **Distributive**: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- **De Morgan's**: $\neg(p \wedge q) \equiv \neg p \vee \neg q$; $\quad \neg(p \vee q) \equiv \neg p \wedge \neg q$
- **Absorption**: $p \vee (p \wedge q) \equiv p$
- **Implication**: $p \to q \equiv \neg p \vee q$
- **Contrapositive**: $p \to q \equiv \neg q \to \neg p$

Worked: prove $\neg(p \to q) \equiv p \wedge \neg q$ by chaining laws

1. KILL THE ARROW FIRST
$\neg(p \to q) \equiv \neg(\neg p \vee q)$ *(Implication law — arrows can't be pushed through directly)*

↓

2. DE MORGAN
$\equiv \neg\neg p \wedge \neg q$

↓

3. DOUBLE NEGATION
$\equiv p \wedge \neg q$ *(name each law per line — that's the graded part)*

↓

4. CROSSCHECK BY TRUTH TABLE
both sides are T only on the row $p=T, q=F$ ✓— two independent proofs, either accepted

Read the result: a conditional fails only when the promise was made and broken — p happened, q didn't.

predicate logic & quantifiers

- **universal** $\forall x\, P(x)$: "for all x, $P(x)$"; true iff P holds for every x.
- **existential** $\exists x\, P(x)$: true iff at least one x satisfies P.
- **uniqueness** $\exists! x\, P(x)$: exactly one x satisfies P.

Negation

- $\neg\forall x\, P(x) \equiv \exists x \neg P(x)$
- $\neg\exists x\, P(x) \equiv \forall x \neg P(x)$
- Through a conditional: $\neg\forall x\,(P(x) \to Q(x)) \equiv \exists x\,(P(x) \wedge \neg Q(x))$ — the counterexample shape: not $\exists x(P \to \neg Q)$

Worked: negate $\forall x \exists y\,(x+y=0)$ step by step

1. PUSH PAST THE $\forall$
$\exists x \neg\exists y\,(x+y=0)$ *(each quantifier flips as $\neg$ passes it)*

↓

2. PUSH PAST THE $\exists$
$\exists x \forall y \neg(x+y=0)$

↓

3. LAND ON THE PREDICATE
$\exists x \forall y\,(x+y \neq 0)$ — "some x has no additive inverse" *(false over $\mathbb{R}$, so the original was true)*

Nested quantifiers

- $\forall x \exists y\, P(x,y)$: each x has some y
- $\exists y \forall x\, P(x,y)$: one y works for all x (stronger)
- $\forall x \forall y \equiv \forall y \forall x$, $\quad \exists x \exists y \equiv \exists y \exists x$
- $\forall x \exists y \not\equiv \exists y \forall x$ (order matters)

The canonical witness over $\mathbb{R}$, $P(x,y)$: $x+y=0$: $\quad \forall x \exists y$ — true (take $y=-x$, a y *per* x); $\quad \exists y \forall x$ — false (no single y zeroes every x). Same predicate, swapped order, opposite verdicts.

Finite-domain evaluation: domain $\{1,2,3\}$, $P(x)$: $x^2>x$: $\quad \forall x\, P(x)$ — false ($x=1$: $1 \not> 1$); $\quad \exists x\, P(x)$ — true ($x=2$). Check every element. One failure kills $\forall$, one witness saves $\exists$.

sets

Unordered collection of distinct **elements**. Order and repetition don't matter: $\{1,2,3\}=\{3,1,2,2\}$.

Common number sets

- $\varnothing=\{\}$: empty set
- $\mathbb{N}$: naturals $\{0,1,2,3,\ldots\}$
- $\mathbb{Z}$: integers $\{\ldots,-1,0,1,\ldots\}$
- $\mathbb{Z}^+$: positive integers
- $\mathbb{Q}$: rationals $\{p/q \mid p,q \in \mathbb{Z}, q \neq 0\}$
- $\mathbb{R}$: real numbers
- $\mathbb{C}$: complex $\{a+bi \mid a,b \in \mathbb{R}\}$

Set-builder notation

$\{x \in S \mid P(x)\}$: "set of x in S such that $P(x)$."

- $\{x \in \mathbb{Z} \mid x>0\}=\mathbb{Z}^+$
- $\{2k \mid k \in \mathbb{Z}\}$ = even integers
- $\{x \in \mathbb{R} \mid x^2<4\}=(-2,2)$

Reverse direction — describe, then build: "positive multiples of 3 under 20" $\to \{3k \mid k \in \mathbb{Z}^+,\ 3k<20\} = \{3,6,\ldots,18\}$ — pick the pattern variable first, then bolt on the conditions.

Membership and subsets

- $x \in A$, $\quad x \notin A$
- $A \subseteq B$: every element of A is in B
- $A \subset B$ or $A \subsetneq B$: $A \subseteq B$ and $A \neq B$

- $A=B$ iff $A \subseteq B$ and $B \subseteq A$
- $\varnothing \subseteq A$ for every A *(vacuously: it has no element that could fail the test)*

The $\in$ vs $\subseteq$ trap, on $A=\{a,b\}$: $a \in A$ but $a \not\subseteq A$ (not a set); $\{a\} \subseteq A$ but also $\{a\} \in \mathcal{P}(A)$ — elements of a power set are subsets of A. Ask "is the left thing an *element* of the right, or a *set of elements* from it?"

Cardinality

$|A|$ = number of elements in A.
- $|\varnothing|=0$, $|\{a,b,c\}|=3$
- **finite** if $|A|=n$ for some $n \in \mathbb{N}$
- **infinite** if not finite
- $|A|=|B|$ iff bijection $A \to B$ exists

Set operations

- **union**: $A \cup B=\{x \mid x \in A \vee x \in B\}$
- **intersection**: $A \cap B=\{x \mid x \in A \wedge x \in B\}$
- **difference**: $A \setminus B=\{x \mid x \in A \wedge x \notin B\}$
- **complement**: $\overline{A}=A^c=U \setminus A$
- **symmetric diff.**: $A \oplus B=(A \setminus B) \cup (B \setminus A)$
- **disjoint**: $A \cap B=\varnothing$

Worked: every operation on one pair

$U=\{1,\ldots,8\}$, $A=\{1,2,3,4\}$, $B=\{3,4,5,6\}$:

1. UNION & INTERSECTION
$A \cup B=\{1,2,3,4,5,6\}$ $\quad$ $A \cap B=\{3,4\}$

↓

2. DIFFERENCES — ORDER MATTERS
$A \setminus B=\{1,2\}$ $\quad$ $B \setminus A=\{5,6\}$

↓

3. COMPLEMENT & SYMMETRIC DIFF
$\overline{A}=\{5,6,7,8\}$ $\quad$ $A \oplus B=\{1,2\} \cup \{5,6\}=\{1,2,5,6\}$ *($\oplus$ = union minus the overlap)*

↓

4. SANITY COUNT
$|A \cup B|=|A|+|B|-|A \cap B|=4+4-2=6$ ✓ — inclusion-exclusion already at work

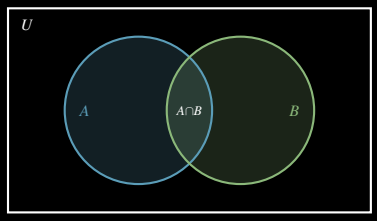
Cartesian product

$A \times B=\{(a,b) \mid a \in A \wedge b \in B\}$
- $(a,b) \neq (b,a)$ in general
- $|A \times B|=|A| \cdot |B|$

Roster example: $\{1,2\} \times \{a,b,c\}=\{(1,a),(1,b),(1,c),(2,a),(2,b),(2,c)\}$ — $2 \cdot 3=6$ pairs, first coordinate from the first set, always. ($\{a,b,c\} \times \{1,2\}$ is a *different* set.)

Power set

$\mathcal{P}(A)$ = all subsets of A (incl. $\varnothing$ and A). $|\mathcal{P}(A)|=2^{|A|}$
Ex: $\mathcal{P}(\{a,b\})=\{\varnothing,\{a\},\{b\},\{a,b\}\}$
Three elements, $2^3=8$ subsets — list by size so none hide:
$\mathcal{P}(\{a,b,c\})=\{\varnothing,\ \{a\},\{b\},\{c\},\ \{a,b\},\{a,c\},\{b,c\},\ \{a,b,c\}\}$. The two usually forgotten: $\varnothing$ and the whole set.



Set identities

- **Identity**: $A \cup \varnothing=A$, $A \cap U=A$
- **Domination**: $A \cup U=U$, $A \cap \varnothing=\varnothing$
- **Idempotent**: $A \cup A=A$, $A \cap A=A$
- **Double comp.**: $\overline{\overline{A}}=A$
- **Commutative**: $A \cup B=B \cup A$
- **Associative**: $(A \cup B) \cup C=A \cup (B \cup C)$
- **Distributive**: $A \cap (B \cup C)=(A \cap B) \cup (A \cap C)$
- **De Morgan's**: $\overline{A \cup B}=\overline{A} \cap \overline{B}$
- **Absorption**: $A \cup (A \cap B)=A$
- **Complement**: $A \cup \overline{A}=U$, $A \cap \overline{A}=\varnothing$
- **Difference**: $A \setminus B=A \cap \overline{B}$

Worked: De Morgan proven both ways

Way 1 — element chasing (double containment in one iff-chain):

1. TAKE AN ARBITRARY ELEMENT
$x \in \overline{A \cup B}$

↓

2. UNPACK TWO DEFINITIONS
complement first: $\iff x \notin A \cup B$; then union: $\iff \neg(x \in A \vee x \in B)$

↓

3. LOGIC DE MORGAN DOES THE WORK
$\iff x \notin A \wedge x \notin B$ *(every set law is a logic law wearing braces)*

↓

4. REPACK
$\iff x \in \overline{A} \cap \overline{B}$ — every step an $\iff$, so both containments at once ■

Way 2 — set algebra: chain known identities, one per step, exactly like the logic-law proof earlier — for example, $\overline{A \cup (B \cap C)}=\overline{A} \cap \overline{B \cap C}=\overline{A} \cap (\overline{B} \cup \overline{C})$ (De Morgan twice). Either style is a full proof. Element-chasing is the fallback when no identity fits.

Inclusion-exclusion

- $|A \cup B|=|A|+|B|-|A \cap B|$
- $|A \cup B \cup C|=|A|+|B|+|C|-|A \cap B|-|A \cap C|-|B \cap C|+|A \cap B \cap C|$
- $|A \setminus B|=|A|-|A \cap B|$
- $|\overline{A}|=|U|-|A|$
- Disjoint: $|A \cup B|=|A|+|B|$

Worked: the 3-set survey

Of 60 students: 30 take Java, 25 Python, 20 C; 12 Java∧Python, 8 Java∧C, 6 Python∧C; 3 all three.

1. AT LEAST ONE — RUN THE FORMULA
$|J \cup P \cup C|=30+25+20-12-8-6+3=52$

↓

2. NONE — COMPLEMENT
$60-52=8$

↓

3. EXACTLY ONE — PEEL THE OVERLAPS PER SET
Java only: $30-12-8+3=13$; Python only: $25-12-6+3=10$; C only: $20-8-6+3=9$ $\Rightarrow$ 32 *(add the triple back — it was subtracted twice)*

functions

$f : A \to B$ assigns each $a \in A$ a unique $f(a) \in B$.
- **domain** A: inputs
- **codomain** B: possible outputs
- **range/image** $f(A)=\{f(a) \mid a \in A\} \subseteq B$: actual outputs

Types

- **injective (1-to-1)**: $f(a_1)=f(a_2) \Rightarrow a_1=a_2$. No two inputs share an output.
- **surjective (onto)**: $\forall b \in B$, $\exists a$ with $f(a)=b$. Range = codomain.
- **bijective**: both injective and surjective. Has inverse.

Worked: prove $f : \mathbb{R} \to \mathbb{R}$, $f(x)=3x+2$ is a bijection

1. INJECTIVE — ASSUME EQUAL OUTPUTS
$3a+2=3b+2 \Rightarrow 3a=3b \Rightarrow a=b$ *(outputs equal forces inputs equal — the definition, verbatim)*

↓

2. SURJECTIVE — CATCH AN ARBITRARY b
given any b, exhibit the input: $a=\frac{b-2}{3}$, and $f(a)=3 \cdot \frac{b-2}{3}+2=b$ ✓

↓

3. CONCLUDE
both hold ⇒ bijective, so f^{-1} exists ■

Disproof by counterexample — $g(x)=x^2$ on $\mathbb{R}\to\mathbb{R}$: not injective ($g(-1)=g(1)=1$, two inputs one output); not surjective (no a with $a^2=-1$). Codomain surgery fixes surjectivity: the same rule as $g:\mathbb{R}\to[0,\infty)$ *is* onto — inj/surj are properties of the whole signature, not the formula.

Composition & inverse

- $(f\circ g)(x)=f(g(x))$
- Associative: $(f\circ g)\circ h=f\circ(g\circ h)$
- Not commutative: $f\circ g\neq g\circ f$ in general
- **Inverse** f^{-1}: exists iff f bijective; $f^{-1}(f(x))=x$
- $(f\circ g)^{-1}=g^{-1}\circ f^{-1}$
- **Identity** $\mathrm{id}_A(x)=x$

Worked: $f(x)=2x+1$, $g(x)=x^2$: $(f\circ g)(x)=2x^2+1$ but $(g\circ f)(x)=(2x+1)^2$ — different functions: non-commutativity on display.
Finding f^{-1} — swap and solve: $y=3x+2\to$ swap: $x=3y+2\to$ solve: $f^{-1}(x)=\frac{x-2}{3}$; check $f^{-1}(f(x))=x$ ✓.

Cardinality comparisons

- Injection $A\to B$: $|A|\le|B|$
- Surjection $A\to B$: $|A|\ge|B|$
- Bijection $A\to B$: $|A|=|B|$
- **Pigeonhole**: if $|A|>|B|$, no injection $A\to B$; *some* output is hit ≥ 2 times.
- **Generalized pigeonhole**: N objects in k boxes ⇒ some box holds $\ge\lceil N/k\rceil$.

Worked: pigeonhole, plain and generalized

1. PLAIN — NAME PIGEONS AND HOLES
13 people, 12 birth months: $13>12$ ⇒ two share a month *(existence only — no telling which month)*

↓

2. GENERALIZED — INVERT THE CEILING
cards to guarantee 3 of one suit: want $\lceil N/4\rceil\ge 3$. The ceiling exceeds 2 exactly when $N/4>2$, that is, $N>8$; smallest integer: $N=9$

↓

3. THE WORST-CASE READ
8 cards can dodge (2 per suit); the 9th card breaks some suit to 3 — "guarantee" always means beat the worst case

relations

A **relation** on A is any set $R\subseteq A\times A$; write aRb for $(a,b)\in R$. Three properties to check, each a quantified statement:

- **reflexive**: $\forall a$: aRa — every element loops to itself
- **symmetric**: $aRb\Rightarrow bRa$ — arrows come in pairs
- **transitive**: $aRb\wedge bRc\Rightarrow aRc$ — paths shortcut

All three ⇒ **equivalence relation**, which partitions A into equivalence classes.

Worked: classify two relations

1. $\le$ ON $\mathbb{Z}$
reflexive ($a\le a$) ✓; symmetric? $2\le 3$ but $3\not\le 2$ fails; transitive ✓— not an equivalence (it's a partial order)

↓

2. $a\equiv b\pmod 3$ ON $\mathbb{Z}$
reflexive ($3\mid a-a=0$) ✓; symmetric ($b-a=-(a-b)$, so $3\mid a-b\Rightarrow 3\mid b-a$) ✓; transitive ($3\mid a-b$, $3\mid b-c\Rightarrow 3\mid(a-b)+(b-c)$, and $(a-b)+(b-c)=a-c$, so $3\mid a-c$) ✓— equivalence

↓

3. READ OFF THE CLASSES
$[0]=\{\ldots,-3,0,3,\ldots\}$, $[1]$, $[2]$ — three classes, disjoint, covering all of $\mathbb{Z}$: a partition *(one class per remainder — mod arithmetic below runs on these)*

number theory

Divisibility

$a\mid b$ ("a divides b"): $\exists k\in\mathbb{Z}$ with $b=ak$.

- $a\mid a$, $a\mid 0$, $1\mid a$
- $a\mid b$ and $a\mid c\Rightarrow a\mid(b+c)$ and $a\mid(bc)$
- $a\mid b$ and $a\mid c\Rightarrow a\mid(mb+nc)$ for any $m,n\in\mathbb{Z}$
- $a\mid b$ and $b\mid c\Rightarrow a\mid c$ (transitive)
- $a\mid b$ and $b\mid a\Rightarrow a=\pm b$

The workhorse, proved (a direct proof in miniature): $a\mid b$, $a\mid c\Rightarrow b=ak$, $c=al\Rightarrow mb+nc=m(ak)+n(al)=a(mk+nl)=a\cdot(\text{integer})\Rightarrow a\mid(mb+nc)$ ■ — unpack, substitute, factor the a out, repack. Every divisibility proof is this move.

Division algorithm

$\forall a\in\mathbb{Z},d\in\mathbb{Z}^+$: $\exists!\,q,r\in\mathbb{Z}$ with $a=dq+r$, $0\le r<d$.

- $q=a$ **div** d: quotient
- $r=a$ **mod** d: remainder

Both signs, worked: $47=5\cdot 9+2$: div $=9$, mod $=2$. Negative dividend: $-11=3\cdot(-4)+1$ — not $3\cdot(-3)-2$: the remainder is never negative; push q down one more.

Base conversion (via div/mod)

Decimal → binary: repeatedly divide by 2, collect remainders, read upward.

Worked: 45 → binary → back

1. DIVIDE DOWN
$45=2\cdot 22+1$, $22=2\cdot 11+0$, $11=2\cdot 5+1$, $5=2\cdot 2+1$, $2=2\cdot 1+0$, $1=2\cdot 0+1$

↓

2. READ REMAINDERS UPWARD
$45=101101_2$ *(bottom remainder is the top bit)*

↓

3. REVERSE DIRECTION — POWERS BACK
$101101_2=32+8+4+1=45$ ✓ *(hex: group bits in 4s from the right: $0010\,1101\to 2D_{16}$)*

Modular arithmetic

$a\equiv b\pmod n$ iff $n\mid(a-b)$ iff a,b have same remainder mod n.

- **Reflexive**: $a\equiv a\pmod n$
- **Symmetric**: $a\equiv b\Rightarrow b\equiv a$
- **Transitive**: $a\equiv b, b\equiv c\Rightarrow a\equiv c$
- **Addition**: $a\equiv b, c\equiv d\Rightarrow a+c\equiv b+d\pmod n$
- **Multiplication**: $\Rightarrow ac\equiv bd\pmod n$
- **Powers**: $a\equiv b\Rightarrow a^k\equiv b^k\pmod n$

Both definitions, one check: $17\equiv 5\pmod{12}$: $12\mid(17-5)=12$ ✓, and both leave remainder 5 ✓.

Worked: the last digit of 7^{100}

Last digit = value mod 10 — the Powers rule's whole job:

1. FIND THE CYCLE
$7^1\to 7$, $7^2\to 9$, $7^3\to 3$, $7^4\to 1$, then repeats — cycle length 4

↓

2. REDUCE THE EXPONENT MOD THE CYCLE
$100\bmod 4=0\Rightarrow 7^{100}$ sits at the *end* of a cycle, like 7^4

↓

3. READ OFF
last digit $=1$ *(exponent $\equiv 0$ means the cycle's last entry, not the first)*

GCD and LCM

- $\gcd(a,b)$: largest d with $d\mid a$ and $d\mid b$
- $\mathrm{lcm}(a,b)$: smallest $m>0$ with $a\mid m$ and $b\mid m$
- **Identity**: $\gcd(a,b)\cdot\mathrm{lcm}(a,b)=|ab|$
- **Coprime**: $\gcd(a,b)=1$

- **Euclidean algorithm**: $gcd(a,b)=gcd(b,a \bmod b)$, stop when second is 0.
- **Bezout**: $\exists s,t \in \mathbb{Z}$ with $sa+tb=gcd(a,b)$.

Worked: Euclid forward — $gcd(252,198)$

1. DIVIDE, KEEP THE REMAINDER
$252=1\cdot198+54$

↓

2. SLIDE LEFT, REPEAT
$198=3\cdot54+36, \qquad 54=1\cdot36+18, \qquad 36=2\cdot18+0$

↓

3. LAST NONZERO REMAINDER
$gcd=18$; bonus: $\text{lcm}=\frac{252\cdot198}{18}=2772$ *(the identity earns its keep)*

Worked: Bézout backward — $18=s\cdot252+t\cdot198$

Run the *same* equations bottom-up, substituting each remainder:

1. START AT THE GCD LINE
$18=54-1\cdot36$

↓

2. REPLACE THE NEWEST REMAINDER
$36=198-3\cdot54$: $\quad 18=54-(198-3\cdot54)=4\cdot54-198$

↓

3. REPLACE AGAIN
$54=252-198$: $\quad 18=4(252-198)-198=4\cdot252-5\cdot198$

↓

4. CHECK
$1008-990=18$ ✓ *($s=4$, $t=-5$ — always verify. Sign slips are the norm here)*

Primes

$p>1$ prime iff only divisors are 1 and p.
- **Fundamental theorem of arithmetic**: every integer >1 has a unique prime factorization.
- Infinitely many primes.
- p prime, $p \mid ab \Rightarrow p \mid a$ or $p \mid b$.

Factorization worked: $600=6\cdot100=2\cdot3\cdot4\cdot25=2^3\cdot3\cdot5^2$ — peel small primes until everything is prime. **Primality by trial division**: test 89: only need primes $\leq \sqrt{89} \approx 9.4$, that is, $2,3,5,7$ — none divide $\Rightarrow$ 89 is prime *(a factor pair straddles $\sqrt{n}$ — one side is always $\leq \sqrt{n}$)*.

Worked: infinitely many primes (by contradiction)

1. ASSUME THE OPPOSITE
finitely many: $p_1, p_2, \ldots, p_n$ is *all* of them

↓

2. BUILD THE BREAKER
$N=p_1p_2\cdots p_n+1$

↓

3. INTERROGATE N
$N>1$, so by the fundamental theorem of arithmetic it has a prime factor q. But each p_i divides $p_1\cdots p_n$, so $N=p_i\cdot(\text{integer})+1$ leaves remainder 1 — $p_i \nmid N$: q is not on the list

↓

4. CONTRADICTION
the list wasn't complete $\Rightarrow$ infinitely many primes ■ *(companion to the $\sqrt{2}$ proof below — same skeleton, different breaker)*

proof strategies

Every proof has the same skeleton: state assumptions, work step-by-step using definitions and prior results, end with the conclusion.

THE PROCESS
1. Read the shape of the claim — $p \to q$, "for all", "there exists", or a set equality.
2. Try to break it first. One counterexample ends the job. Do this before proving anything.
3. Pick the technique from the table below.
4. Write the first and last lines before the middle — what you assume, and what you must reach.
5. Close the gap between them.

Step 3 decoded — which technique?

technique	reach for it when
Direct	p unpacks cleanly into q
Contrapositive	$\neg q$ is easier to handle than p — often when q says "not", or denies existence
Contradiction	irrationality, infinitude, uniqueness
Cases	a definition splits: absolute value, parity, sign
Induction	the claim is indexed by $\mathbb{N}$ and the pattern propagates
Strong induction	recurrences, or splitting into smaller pieces
Counterexample	you suspect the claim is false

THE TWO LINES THAT DO THE WORK
Opening — unpack the hypothesis into algebra: "n is even, so $n=2k$ for some integer k."
Closing — state the goal in the same algebra: "so $n^2=2m$ for some integer m, and n^2 is even."
Write both before the middle. Most stuck proofs are stuck because the goal was never written down.

Direct proof (for $p \to q$)

1. **Assume** p is true.
2. **Apply definitions**: unpack what p means (for example, "n is even" $\Rightarrow n=2k$ for some $k \in \mathbb{Z}$).
3. **Manipulate** algebraically toward the form of q.
4. **Conclude** q is true. Therefore $p \to q$.

Worked: if n is even, then n^2 is even

1. ASSUME p — UNPACK THE DEFINITION
n even: $n=2k$ for some $k \in \mathbb{Z}$

↓

2. MANIPULATE
$n^2=(2k)^2=4k^2$

↓

3. REPACK INTO q'S SHAPE
$=2(2k^2)=2\cdot(\text{integer})$

↓

4. CONCLUDE
n^2 is even ■ *(the template proof — every direct proof is unpack → algebra → repack)*

Contrapositive (for $p \to q$)

Use when assuming p directly is awkward but $\neg q$ gives a clean handle. Valid since $p \to q \equiv \neg q \to \neg p$.
1. **Assume** $\neg q$.
2. Derive $\neg p$ via direct proof.
3. Conclude $p \to q$.

Quick: If n^2 is odd, then n is odd — contrapositive: if n even, n^2 even, proved above, done ■.

Worked: if x^3 is irrational, then x is irrational

1. FLIP TO THE CONTRAPOSITIVE
prove instead: x rational $\Rightarrow x^3$ rational *("irrational" has no algebra handle; "rational" does)*

↓

2. UNPACK
$x=\frac{a}{b}, \quad a,b \in \mathbb{Z}, b \neq 0$

↓

3. CUBE
$x^3=\frac{a^3}{b^3}$ — integers over a nonzero integer: rational

↓

4. CONCLUDE THE ORIGINAL
contrapositive proved $\Rightarrow$ original holds ■

Contradiction (for p)

1. **Assume** $\neg p$. (For a *conditional* $p \to q$, assume $p \wedge \neg q$, since $\neg(p \to q) \equiv p \wedge \neg q$.)
2. Derive a contradiction $q \wedge \neg q$ (something impossible).
3. Conclude $\neg p$ is false, so p is true.

Worked: $\sqrt{2}$ is irrational — the canonical contradiction

1. ASSUME THE OPPOSITE, IN LOWEST TERMS
$\sqrt{2}=\frac{a}{b}$ with $gcd(a,b)=1$ *(the "lowest terms" clause is load-bearing — it's what breaks)*

↓

2. SQUARE & CLEAR
square both sides: $2=\frac{a^2}{b^2}$; multiply by b^2: $2b^2=a^2$

↓

3. SO a^2 IS EVEN, HENCE a IS EVEN
$a^2=2b^2=2\cdot$(integer) — even. And odd numbers have odd squares: $a=2m+1 \Rightarrow a^2=4m^2+4m+1=2(2m^2+2m)+1$, odd — so a^2 even forces a even: $a=2k$

↓

4. SUBSTITUTE — SO b IS EVEN TOO
$a=2k$: $2b^2=(2k)^2=4k^2 \Rightarrow b^2=2k^2$ — even, and by the same lemma b is even

↓

5. CONTRADICTION
both even $\Rightarrow gcd(a,b) \geq 2$ — but we said lowest terms. Assumption dies: $\sqrt{2}$ irrational ■ *(same skeleton proves $\sqrt{3}$)*

Conditional flavor (assume $p \wedge \neg q$): claim: if $3n+2$ is odd then n is odd. Assume $3n+2$ odd *and* n even: $n=2k \Rightarrow 3n+2=6k+2=2(3k+1)$ — even, contradicting the assumption ■.

True/false — try to break it first

- Throw specific numbers at it (especially $\sqrt{2}$). Breaks $\Rightarrow$ false, give that counterexample.
- Can't break it $\Rightarrow$ likely true, write the proof.
- $\sqrt{2}$ as a tool: $(\sqrt{2})^2=2$ (rational!) kills any "irrational $\Rightarrow$ irrational after squaring" claim.

Worked: a disproof, written up properly

Claim: "n^2+n+41 is prime for every $n \in \mathbb{N}$."

1. HUNT A BREAKER
primes for $n=0,1,2,\ldots$ — suspiciously good. Rewrite: $n^2+n+41=n(n+1)+41$ — a 41 will factor out whenever $41 \mid n(n+1)$, for example, $n=40$ or $n=41$

↓

2. EXHIBIT THE COUNTEREXAMPLE
$n=40$: $40\cdot41+41=41(40+1)=41^2=1681$ — not prime *($n=41$ works too: $41\cdot42+41=41\cdot43$)*

↓

3. CONCLUDE
the claim is false; the counterexample *is* the complete answer ■ *(one failure kills a $\forall$ — no further explanation owed)*

Proof by cases

Used when the domain naturally splits.

1. Identify cases that exhaust all possibilities (every input falls into at least one).
2. Prove the statement for each case independently.
3. Conclude it holds in general.

Worked: $|x| \geq 0$ for all real x — and the parity classic

1. SPLIT WHERE THE DEFINITION SPLITS
$|x|$ is piecewise at 0: cases $x \geq 0$ and $x<0$ exhaust $\mathbb{R}$

↓

2. CASE 1
$x \geq 0$: $|x|=x \geq 0$ ✓

↓

3. CASE 2
$x<0$: $|x|=-x>0$ ✓

↓

4. BOTH ROADS ARRIVE
$|x| \geq 0$ always ■

Parity flavor (the usual exam split): n^2+n is even for every n:
$n^2+n=n(n+1)$ — consecutive integers, one is always even ■ *(or: case n even, case n odd — both give even).*

Universal $\forall x\, P(x)$

Let x be **arbitrary** (don't pick a specific value); show $P(x)$ holds. Conclude it holds for all x. *One counterexample disproves a universal claim.*
Worked: "the average of any two distinct reals lies strictly between them": let $a<b$ be arbitrary; then $a=\frac{a+a}{2}<\frac{a+b}{2}<\frac{b+b}{2}=b$ ■ — no specific numbers ever appeared. That's what makes it universal.

Existential $\exists x\, P(x)$

Constructive: exhibit a specific x that works. **Non-constructive**: prove x must exist without finding it (for example, pigeonhole, intermediate value).
Worked (constructive): "some perfect square is a sum of two positive perfect squares": $25=9+16$ ■ — the witness is the whole proof.
Non-constructive: the 13-birthdays pigeonhole above proves two people share a month without naming the month.

Set proofs

- **Subset** $A \subseteq B$: let $x \in A$ arbitrary. Show $x \in B$.
- **Equality** $A=B$: double containment — prove $A \subseteq B$ *and* $B \subseteq A$.
- **Identity** (for example, De Morgan's for sets): prove $x \in$ LHS $\iff x \in$ RHS via logical equivalences — fully worked in the De Morgan chain back in the sets section (element-chasing + law-algebra, both directions in one $\iff$ chain).

Common pitfalls

- Assuming what you're trying to prove (circular).
- Working from the conclusion backward without justifying each step is reversible.
- Using a specific value when $\forall$ requires arbitrary.
- Forgetting to check both directions in iff or set equality.
- Skipping the "by definition" unpacking step.

mathematical induction

Proves $P(n)$ for all integers $n \geq n_0$ via a chain reaction: if it holds at one value, it holds at the next. Like dominos — knock the first, ensure each falls onto the one after it.

Proof structure (template)

1. **State** $P(n)$: write the claim being proved as a function of n.
2. **Base case**: verify $P(n_0)$ directly (usually $n_0=0$ or 1). Plug in, show LHS = RHS.
3. **Inductive hypothesis (IH)**: *assume* $P(k)$ holds for arbitrary $k \geq n_0$. Write the full equation explicitly.
4. **State the goal**: write $P(k+1)$ by substituting $k+1$ for every n in the formula.
5. **Inductive step**: start from LHS of $P(k+1)$, apply the IH, simplify to RHS of $P(k+1)$.
6. **Conclude**: "By the principle of mathematical induction, $P(n)$ holds for all integers $n \geq n_0$." ■

The key move: applying the IH

LHS of $P(k+1)$=[entire sum up to k]+[new $(k+1)$-th term]. The IH replaces the *entire sum*, not just the last term:

$$\underbrace{a_1+a_2+\cdots+a_k}_{\text{replaced via IH}}+a_{k+1} \quad \rightarrow \quad [\text{formula at } k]+a_{k+1}$$

Then it's algebra: combine, factor, and aim for the RHS of $P(k+1)$. Always work LHS $\rightarrow$ RHS — never the reverse, since you can't assume the conclusion.

Substituting $n \to k+1$

Replace *every* n in the formula. Wrap in parentheses to avoid sign errors:

- $n \to (k+1)$
- $n+1 \to (k+1)+1 = k+2$
- $2n+1 \to 2(k+1)+1 = 2k+3$
- $\frac{n(n+1)}{2} \to \frac{(k+1)(k+2)}{2}$
- $2^n \to 2^{k+1}$, $(-1)^n \to (-1)^{k+1}$

Keep the result factored where possible — don't expand unless you're matching it to a specific polynomial form.

Algebra toolkit for the inductive step

Almost every induction step reduces to one of these moves:

- **Common denominator**: $\frac{A}{b^k} + B = \frac{A + Bb^k}{b^k}$
- **Factor out shared term**: $\frac{k(k+1)}{2} + (k+1) = (k+1)\left[\frac{k}{2}+1\right] = \frac{(k+1)(k+2)}{2}$
- **Combine same-base exponents**: $2 \cdot 2^k = 2^{k+1}$
- **Recognize the target**: keep an eye on RHS of $P(k+1)$ — your steps should be aiming there.

Worked: sum formula — $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$

1. STATE $P(n)$
$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$, claim for all $n \geq 1$

↓

2. BASE CASE $P(1)$
LHS $= 1$; RHS $= \frac{1 \cdot 2}{2} = 1$ ✓

↓

3. IH — ASSUME $P(k)$
$\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$ *(write the full equation — it's the tool you'll swing)*

↓

4. GOAL — WRITE $P(k+1)$
$\sum_{i=1}^{k+1} i \stackrel{?}{=} \frac{(k+1)(k+2)}{2}$

↓

5. STEP — PEEL, APPLY IH
$$\sum_{i=1}^{k+1} i = \underbrace{\sum_{i=1}^{k} i}_{\text{IH}} + (k+1) = \frac{k(k+1)}{2} + (k+1)$$

↓

6. FACTOR TO THE TARGET
$= (k+1)\left[\frac{k}{2}+1\right] = (k+1) \cdot \frac{k+2}{2} = \frac{(k+1)(k+2)}{2}$ ✓ — by induction, $P(n)$ holds for all $n \geq 1$ ■

Common pitfalls

- Replacing only the last term, not the whole sum up to k.
- Treating "…" as if those terms don't exist — they do.
- Confusing $\sum$ with individual terms (sigma gives *partial sums*).
- Wrong base case: $P(0)$ for "all nonnegative integers", $P(1)$ for "all positive integers".
- Working RHS backward to LHS (circular).
- Forgetting to wrap $k+1$ in parentheses when substituting.

Worked: the other three flavors

Same six moves. Only step 5's algebra changes. One chain per flavor:

Divisibility — $6 \mid n^3 - n$:

1. BASE $P(1)$
$1 - 1 = 0$, and $6 \mid 0$ ✓

↓

2. STEP — EXPAND, THEN REGROUP AROUND THE IH
$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - k - 1 = \underbrace{(k^3 - k)}_{\text{IH: } 6\mid} + 3k(k+1)$$

↓

3. HANDLE THE LEFTOVER
$k(k+1)$ is consecutive $\Rightarrow$ even: $k(k+1) = 2m$, so $3k(k+1) = 6m$: sum of two multiples of 6 ■ *(the move: expand, then regroup until the IH's expression appears)*

Inequality — $2^n > n^2$ for $n \geq 5$:

1. BASE $P(5)$ — NOT $P(1)$!
$32 > 25$ ✓ *(true at $n=1$ but false at $n=2,3,4$ — base at the first n where it's true and stays true, that is, where the step below can carry it)*

↓

2. STEP — DOUBLE THE IH
$2^{k+1} = 2 \cdot 2^k > 2k^2 = k^2 + k^2$

↓

3. BOUND THE SPARE k^2
$k \geq 5$: $k^2 \geq 5k = 2k + 3k$, and $3k \geq 15 \geq 1$, so $k^2 \geq 2k+1$. Feed it back into step 2: $2^{k+1} > k^2 + k^2 \geq k^2 + 2k + 1 = (k+1)^2$ ■ *(inequalities chain loosely — you only need $\geq$, not $=$)*

Geometric — $\sum_{i=0}^{n} 2^i = 2^{n+1} - 1$:

1. BASE $P(0)$
$2^0 = 1 = 2^1 - 1$ ✓

↓

2. STEP — PEEL THE LAST TERM, APPLY THE IH
$$\sum_{i=0}^{k+1} 2^i = \left(\sum_{i=0}^{k} 2^i\right) + 2^{k+1} \stackrel{\text{IH}}{=} (2^{k+1} - 1) + 2^{k+1}$$

↓

3. COMBINE BASES
$= 2^{k+1} + 2^{k+1} - 1 = 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1$ ■ *($b \cdot b^k = b^{k+1}$ finally earns its keep)*

Fractions — $\sum_{i=1}^{n} \frac{1}{i(i+1)} = \frac{n}{n+1}$: base $\frac{1}{1 \cdot 2} = \frac{1}{2}$ ✓. Step: peel and apply the IH, then it's exactly the common-denominator case worked in the algebra refresher:

$$\sum_{i=1}^{k+1} \frac{1}{i(i+1)} \stackrel{\text{IH}}{=} \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} = \frac{k(k+2)+1}{(k+1)(k+2)}$$

$$= \frac{k^2+2k+1}{(k+1)(k+2)} = \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2} \blacksquare$$

Conjecturing a formula

When you have to find the formula yourself before proving it:

1. Compute **partial sums** for $n = 1, 2, 3, 4$ — running totals, not individual terms.
2. Look at what the sums *equal* — that's where the pattern lives.
3. Try simplified *and* unsimplified forms ($\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ vs $\frac{1}{2}, \frac{4}{6}, \frac{18}{24}$).
4. Break numerator and denominator patterns apart.
5. Test conjecture at an unused n before committing to the proof.

Worked: conjecture, then prove

For $\sum_{i=1}^{n} \frac{1}{i(i+1)}$:

1. PARTIAL SUMS
$\frac{1}{2}$, $\frac{1}{2} + \frac{1}{6} = \frac{2}{3}$, $+\frac{1}{12} = \frac{3}{4}$, $+\frac{1}{20} = \frac{4}{5}$

↓

2. SPOT THE PATTERN
$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}$: conjecture $\frac{n}{n+1}$

↓

3. TEST AN UNUSED n
$n = 5$: $\frac{4}{5} + \frac{1}{30} = \frac{25}{30} = \frac{5}{6}$ ✓

↓

4. NOW PROVE IT
the fraction-flavor induction above *(conjecture → spot-check → induct: the full workflow)*

Strong induction

Same skeleton, but the IH assumes $P(n_0), P(n_0+1), \ldots, P(k)$ *all* hold (not just $P(k)$). Use when $P(k+1)$ depends on multiple earlier values:

- Recurrences like Fibonacci (F_{k+1} depends on F_k and F_{k-1}).
- Prime factorization (each factor is smaller, but not necessarily k).

- "Every integer >1 has property X" when the inductive step splits the integer.

The base case may need multiple values verified (for example, , both $P(0)$ and $P(1)$ for Fibonacci).

Worked: postage stamps — every $n \geq 8$ from 3's and 5's

1. BASES — ONE PER REACH-BACK
$8=3+5$, $9=3+3+3$, $10=5+5$ ✓ *(the step reaches back 3, so three consecutive bases are needed)*

↓

2. STRONG IH
assume *every* value from 8 through k is makeable ($k \geq 10$)

↓

3. STEP — REACH BACK
$(k+1)-3=k-2 \geq 8$, so $k-2$ is covered by the IH

↓

4. ADD ONE STAMP
its combination + one 3-stamp makes $k+1$ ■ *(ordinary induction can't do this — $k-2$ isn't k)*

Discrete II — Counting, Recurrences & Graphs

sequences & recurrences

Sequence types

- **Arithmetic**: constant difference d. $a_n=a_1+(n-1)d$. Sum: $S_n=\frac{n(a_1+a_n)}{2}$.
- **Geometric**: constant ratio r. $a_n=a_1 \cdot r^{n-1}$. Sum: $S_n=a_1\frac{r^n-1}{r-1}$ ($r \neq 1$).
- **Infinite geometric** $|r|<1$: $\sum_{n=0}^{\infty} ar^n=\frac{a}{1-r}$.

Worked: arithmetic $a_1=7$, $d=4$: $a_{20}=7+19(4)=83$, $S_{20}=\frac{20(7+83)}{2}=900$. **From two terms**: $a_3=11$, $a_8=26$: five steps apart, so $a_8-a_3=5d$: $26-11=15 \Rightarrow d=3$; back up two steps: $a_1=a_3-2d=11-6=5$. **Infinite**: $\sum_{n=0}^{\infty} 3(\frac{1}{4})^n=\frac{3}{1-1/4}=4$ *(converges only because $|r|<1$)*.

Recurrence relations

Define a_n in terms of earlier terms plus initial conditions.

- **Linear**: $a_n=c_1a_{n-1}+c_2a_{n-2}+\cdots$
- **Homogeneous**: no constant/input term.
- Ex: Fibonacci $F_n=F_{n-1}+F_{n-2}$, $F_0=0, F_1=1$.

Worked: unroll → guess → prove

$a_n=2a_{n-1}+1$, $a_0=1$:

1. UNROLL A FEW
1, 3, 7, 15, 31

↓

2. GUESS
each is one below a power of two: $a_n=2^{n+1}-1$

↓

3. PROVE BY INDUCTION
base: $a_0=1=2^1-1$ ✓; IH: $a_k=2^{k+1}-1$; step: $a_{k+1}=2a_k+1 \overset{\text{IH}}{=} 2(2^{k+1}-1)+1=2^{k+2}-2+1=2^{k+2}-1$ ✓ *(unrolling conjectures. Induction certifies — the workflow from Discrete I)*

Solving linear homogeneous (2nd order)

$a_n=c_1a_{n-1}+c_2a_{n-2}$:

1. Characteristic equation: $r^2-c_1r-c_2=0$. Solve for roots r_1, r_2.
2. Distinct roots: $a_n=Ar_1{}^n+Br_2{}^n$.
3. Repeated root r: $a_n=(A+Bn)r^n$.
4. Use initial conditions to solve for A, B.

Worked (distinct roots): $a_n=a_{n-1}+6a_{n-2}$, $a_0=3$, $a_1=4$

1. CHARACTERISTIC EQUATION
$r^2-r-6=0$ *(coefficients cross with signs flipped)*

↓

2. ROOTS
$(r-3)(r+2)=0$: $r=3, -2$ — distinct

↓

3. GENERAL FORM + INITIAL CONDITIONS
$a_n=A\,3^n+B(-2)^n$: $n=0$: $A+B=3$; $n=1$: $3A-2B=4$

↓

4. SOLVE THE 2×2
from the first: $B=3-A$; substitute: $3A-2(3-A)=4 \Rightarrow 5A-6=4 \Rightarrow A=2$, so $B=1$: $a_n=2\cdot 3^n+(-2)^n$

↓

5. CHECK a_2 BOTH WAYS
recurrence: $4+18=22$; formula: $18+4=22$ ✓

Worked (repeated root): $a_n=4a_{n-1}-4a_{n-2}$, $a_0=1$, $a_1=4$

1. CHARACTERISTIC
$r^2-4r+4=(r-2)^2$: $r=2$ doubled

↓

2. REPEATED-ROOT FORM
$a_n=(A+Bn)2^n$ *(the extra n is what a double root buys — $A2^n+B2^n$ alone can't fit two conditions)*

↓

3. INITIAL CONDITIONS
$n=0$: $A=1$; $n=1$: $(1+B)2=4 \Rightarrow B=1$: $a_n=(1+n)2^n$

↓

4. CHECK
a_2: recurrence $16-4=12$; formula $3\cdot 4=12$ ✓

Fibonacci runs the distinct-roots case with $r=\frac{1\pm\sqrt{5}}{2}$ — same machine, uglier numbers (Binet's formula).

combinatorics

Counting rules

- **Sum rule** (OR, disjoint): A or B: $|A|+|B|$ ways. Triggers: "or," "either."
- **Product rule** (AND, sequential): A then B: $|A| \cdot |B|$ ways. Triggers: "and," "then."
- **Complement**: $|A|=|U|-|\overline{A}|$. Use for "at least one" → count the opposite "none." If each of n slots has k non-X options, the complement is k^n.

THE PROCESS — TWO QUESTIONS DECIDE EVERYTHING
1. Does order matter? Swap two chosen items — does the result change?
2. Is repetition allowed? May an item be picked twice?
3. Read the count off the table.
4. Check for overlap or symmetry — then correct with inclusion-exclusion or the division rule.

Steps 1–2 decoded — the counting table

	no repetition	**repetition allowed**
order matters	$P(n,k)=\frac{n!}{(n-k)!}$	n^k
order does not	$C(n,k)=\binom{n}{k}$	$\binom{n+k-1}{k}$

The bottom-right cell is **stars and bars**. Link: $P(n,k)=C(n,k)\cdot k!$ — a permutation is a combination times its orderings.

Step 4 decoded — three corrections

A **"at least one"** ⇒ count the complement and subtract from the total. Counting it directly means counting every case.
B **overlapping sets** ⇒ inclusion-exclusion: $|A \cup B|=|A|+|B|-|A \cap B|$
C **identical items or a symmetry** ⇒ division rule: count as if distinct, then divide by the duplicates.

Worked: "at least one" — 8-bit strings with at least one 0

1. COUNT EVERYTHING
$2^8=256$ strings

↓

2. COUNT THE OPPOSITE — "NO ZEROS"
all ones: exactly 1 string

↓

3. SUBTRACT
$256-1=255$ *(counting "at least one" directly means summing 8 overlapping cases — the complement is one subtraction)*

Inclusion-exclusion (overlapping sets)

$$|A\cup B|=|A|+|B|-|A\cap B|$$

$$|A\cup B\cup C|=|A|+|B|+|C|-|A\cap B|-|A\cap C|-|B\cap C|+|A\cap B\cap C|$$

Pattern: $+$ singles $-$ pairs $+$ triples $-\cdots$ Fully worked (3-set survey: 52/8/32) in the Discrete I sets section — same formula, don't relearn it.

Division rule (kill duplicates)

$$\text{true count}=\frac{\text{overcount}}{\text{duplicates per item}}$$

- **Circular** arrangements: $\frac{n!}{n}=(n-1)!$ (rotations are the same).
- **Indistinguishable** objects: $\frac{n!}{n_1!n_2!\cdots}$ (swapping identical items doesn't count).

Worked: both division flavors

1. ROUND TABLE, 6 PEOPLE
line them up: $6!=720$; every seating counted 6 times (rotations): $\frac{720}{6}=5!=120$

↓

2. MISSISSIPPI
11 letters: $11!$; identical letters over-count: I$\times$4, S$\times$4, P$\times$2: $\frac{11!}{4!4!2!}=34{,}650$ *(one factorial in the denominator per repeated letter)*

Permutations (order matters)

Ordered arrangements of k from n:

$$P(n,k)=\frac{n!}{(n-k)!}=n(n-1)(n-2)\cdots(n-k+1)$$

Full permutation: $P(n,n)=n!$. $\quad 0!=1$.

Combinations (order doesn't matter)

k-subsets of an n-set:

$$C(n,k)=\binom{n}{k}=\frac{n!}{k!(n-k)!}$$

- $\binom{n}{0}=\binom{n}{n}=1$, $\binom{n}{1}=n$
- **Symmetry**: $\binom{n}{k}=\binom{n}{n-k}$
- **Pascal**: $\binom{n}{k}=\binom{n-1}{k-1}+\binom{n-1}{k}$

Worked: choose 5 from 10 — both ways at once

1. ORDER MATTERS (ranked top-5)
$P(10,5)=10\cdot9\cdot8\cdot7\cdot6=30{,}240$

↓

2. ORDER DOESN'T (a team of 5)
$C(10,5)=\frac{10!}{5!5!}=252$

↓

3. THE LINK
$\frac{30{,}240}{252}=120=5!$ ✓— a permutation is a combination times its orderings, $P=C\cdot k!$

Binomial theorem

$$(x+y)^n=\sum_{k=0}^{n}\binom{n}{k}x^{n-k}y^k$$

- $\sum_{k=0}^{n}\binom{n}{k}=2^n$ (all subsets) — *(substitute $x=y=1$)*
- $\sum_{k=0}^{n}(-1)^k\binom{n}{k}=0$ $(n\geq 1)$ — *(substitute $x=1$, $y=-1$)*

Worked: extract one coefficient

Coefficient of x^5y^3 in $(x+y)^8$: the y^k exponent names the term: $k=3$: $\binom{8}{3}=56$.

Coefficient of x^3 in $(2x-3)^5$:

1. NAME THE TERM
x^3 needs $k=2$: $\binom{5}{2}(2x)^3(-3)^2$

↓

2. MULTIPLY *ALL* THE CONSTANTS
$10\cdot8\cdot9=720$ *(the 2^3 and $(-3)^2$ are part of the coefficient — quoting $\binom{5}{2}=10$ alone is the classic loss)*

Repetition allowed

- **Sequences w/ repetition**: n^k (each of k slots from n options)
- **Multisets of size k from n**: $\binom{n+k-1}{k}$ (stars-and-bars)

Worked: stars and bars — 6 donuts, 4 types

1. MODEL
order irrelevant, repetition allowed: a multiset of size $k=6$ from $n=4$

↓

2. THE PICTURE
6 stars split by $n-1=3$ bars: $\star\star\mid\star\mid\star\star\star\mid$ means $2,1,3,0$ of each type

↓

3. COUNT THE ARRANGEMENTS
9 symbol slots. Choose the 3 bar positions (equivalently the 6 star positions — symmetry): $\binom{9}{3}=\binom{9}{6}=\binom{6+4-1}{6}=84$

The full 2×2 grid: order+no-rep $\to P(n,k)$; no-order+no-rep $\to C(n,k)$; order+rep $\to n^k$; no-order+rep $\to$ stars and bars.

Permutation vs combination?

Ask: *"If I swap two chosen items, does the result change?"* Yes $\to$ permutation (order matters); No $\to$ combination.

Scenario	Type
Race ranks, 1st/2nd/3rd	P
Named roles (Pres/VP)	P
Left-to-right arrangement	P
Committee / team (no roles)	C
Poker hand, subset	C

Link: $P(n,k)=C(n,k)\cdot k!$ — a permutation is a combination times its orderings. Calc tip: parenthesize the *whole* denominator, else left-to-right gives wrong answers.

asymptotic notation

Describes long-term growth of functions. For $f,g:\mathbb{N}\to\mathbb{R}^+$:

- $f=O(g)$: f grows *no faster than* g. $\quad\exists c,n_0$: $f(n)\leq cg(n)\,\forall n\geq n_0$.
- $f=\Omega(g)$: f grows *no slower than* g. $\quad\exists c,n_0$: $f(n)\geq cg(n)\;\forall n\geq n_0$.
- $f=\Theta(g)$: f grows *at same rate as* g. Both O and Ω.
- $f=o(g)$: f grows *strictly slower* than g. $\lim f/g=0$.
- $f=\omega(g)$: f grows *strictly faster*. $\lim f/g=\infty$.

Growth hierarchy (slow $\to$ fast)

$$1\ll\log\log n\ll\log n\ll\sqrt{n}\ll n\ll n\log n\ll n^2\ll n^3\ll 2^n\ll n!\ll n^n$$

Rules of thumb

- Drop constants: $O(3n^2)=O(n^2)$.
- Keep dominant term: $O(n^2+n)=O(n^2)$.
- $\log_a n=\Theta(\log_b n)$ (base doesn't matter in O).
- Polynomial $\ll$ exponential: $n^k=o(c^n)$ for $c>1$.
- $O(f)+O(g)=O(max(f,g))$.
- $O(f)\cdot O(g)=O(fg)$.

Worked: prove $3n^2+5n+2=O(n^2)$ from the definition

1. BOUND EVERY TERM BY THE BIGGEST
for $n\geq 1$: $5n\leq 5n^2$ and $2\leq 2n^2$, so $3n^2+5n+2\leq 10n^2$

↓

2. EXHIBIT THE WITNESSES
$c=10$, $n_0=1$: definition satisfied ■ *(witnesses aren't unique — $c=4, n_0=6$ also works. Any pair closes the proof)*

↓

3. THE LOWER BOUND IS FREE
$3n^2+5n+2\geq 3n^2$ always: $\Omega(n^2)$ with $c=3\Rightarrow$ both bounds $\Rightarrow \Theta(n^2)$

Disproof: $n^2\not\in O(n)$ — given *any* c, n_0, take $n>max(c,n_0)$: then $n^2=n\cdot n>cn$ — no witnesses can exist ■.

Worked: classify code

Nested loops, each running to n, constant body: $n\cdot n=\Theta(n^2)$. Halving loop (`while n>1: n=n/2`): $\Theta(\log n)$ steps. Loop to n containing a halving loop: $\Theta(n\log n)$ — multiply nested, add sequential.

graph theory basics

Graph $G=(V,E)$: vertices V, edges $E\subseteq V\times V$.

- **Undirected**: edges unordered pairs $\{u,v\}$.
- **Directed (digraph)**: edges ordered (u,v).
- **Simple**: no loops, no multi-edges.
- **Weighted**: edges carry numbers.

Degree

- $deg(v)$: number of edges incident to v (undirected).
- **In-deg, out-deg** for directed.
- **Handshake**: $\sum_{v\in V} deg(v)=2|E|$.
- Corollary: number of odd-degree vertices is even.

Worked: handshake, all three directions

1. DEGREES → EDGES
10 vertices, each degree 3: $|E|=\frac{10\cdot 3}{2}=15$

↓

2. EDGES → A MISSING DEGREE
$|E|=7$ and five vertices have degrees $2,2,3,4,?$: $\sum deg=2(7)=14$; known sum $2+2+3+4=11$; missing $=14-11=3$

↓

3. FEASIBILITY VETO
"a graph with degrees $3,3,3,3,3$"? sum $=15$, odd — impossible: odd-degree vertices must come in pairs

Paths & connectivity

- **Walk**: vertex/edge sequence, repetition allowed.
- **Path**: walk with distinct vertices.
- **Cycle**: closed path (start = end), length ≥ 3.
- **Connected**: path between every pair of vertices.
- **Component**: maximal connected subgraph.

Special graphs

- K_n: **complete**, every pair adjacent. $|E|=\binom{n}{2}$.
- C_n: **cycle** on n vertices.
- P_n: **path** on n vertices.
- $K_{m,n}$: **complete bipartite**, two parts of sizes m,n.
- **Bipartite**: $V=V_1\cup V_2$ disjoint, edges only across. Iff no odd cycles.

Examples: $|E(K_5)|=\binom{5}{2}=10$ — combinatorics cashing in. **Bipartite check**: C_6 — alternate 2-color around the even cycle, no clash $\Rightarrow$ bipartite; C_5 — the fifth vertex must match its neighbor $\Rightarrow$ odd cycle $\Rightarrow$ not bipartite. Two-color greedily. A forced clash *is* your odd cycle.

Trees

Connected graph with no cycles.

- Tree on n vertices has exactly $n-1$ edges.
- Unique path between any two vertices.
- **Rooted tree**: one vertex designated as root.
- **Leaf**: vertex of degree 1 (non-root).
- **Binary tree**: each node has ≤ 2 children.
- Adding any edge creates exactly one cycle.

Count check: a connected graph on 7 vertices with 6 edges and no cycle *is* a tree. With 7 edges it has exactly one cycle — edges $=n-1$ is the tree signature.

Adjacency matrix & list

Two standard graph representations — square matrix A with $A_{ij}=1$ iff edge $\{i,j\}$, or a per-vertex neighbor list. **Worked** — the triangle-plus-tail 1–2, 2–3, 3–1, 3–4:

$$A=\begin{pmatrix}0&1&1&0\\1&0&1&0\\1&1&0&1\\0&0&1&0\end{pmatrix}\qquad \begin{matrix}1:2,3\\2:1,3\\3:1,2,4\\4:3\end{matrix}$$

Row sums = degrees $(2,2,3,1$ — handshake: sum $8=2|E|$, $|E|=4$ ✓); undirected $\Rightarrow$ symmetric matrix, zero diagonal (no loops).

m-ary tree math

(m-ary = every node has $\leq m$ children.) These counts need a **full** m-ary tree (every internal node has *exactly* m children). With i internal nodes, l leaves, v vertices, e edges:

- Leaves: $l=(m-1)i+1$
- Vertices: $v=mi+1$, and $v=l+i$
- Edges: $e=v-1=mi$

Perfect tree (full *and* every level filled):

- Nodes at level k: m^k; total through height h: $\frac{m^{h+1}-1}{m-1}$
- Height from l leaves: $h=\lceil \log_m l\rceil$

Worked: m-ary counts, both directions

1. GIVEN INTERNAL, FIND LEAVES
full 3-ary, $i=100$ internal: $l=(3-1)100+1=201$ leaves; $v=3(100)+1=301$ *(check: $301=201+100$ ✓)*

↓

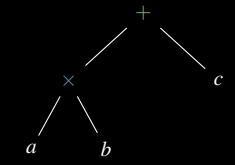
2. GIVEN LEAVES, FIND INTERNAL — SOLVE BACKWARD
full 3-ary with $l=100$: $100=2i+1\Rightarrow i=49.5$ — not an integer $\Rightarrow$ impossible; l must be odd for $m=3$. With $l=101$: $i=50$

↓

3. HEIGHT BOUND
perfect binary with $l=64$ leaves: $h=\log_2 64=6$ *(the formulas run in every direction — treat each as an equation, not a recipe)*

Expression trees

- **Internal nodes** = operators; **leaves** = numbers/variables.
- The root is the *last* operation done (reverse PEMDAS).
- Low-priority operators sit high; high-priority sit deep.



tree for $a \times b + c$

Worked: build the tree from $(a+b)\times c-d$

1. LAST OPERATION = ROOT
reverse PEMDAS: the $-$ happens last $\Rightarrow$ root $-$

↓

2. RECURSE ON EACH SIDE
left subtree: $(a+b)\times c$ — its last op is $\times$; right subtree: leaf d

↓

3. KEEP SPLITTING
$\times$'s children: the parenthesized $+$ (with leaves a, b) and leaf c — parentheses force $+$ deeper than its usual priority

↓

4. READ OFF THE NOTATIONS
postorder: $ab+c\times d-$; preorder: $-\times+abcd$

Tree traversals = notation

- **Preorder** (Root, Left, Right) $\rightarrow$ prefix: $+\times abc$
- **Inorder** (Left, Root, Right) $\rightarrow$ infix (normal math): $a\times b+c$
- **Postorder** (Left, Right, Root) $\rightarrow$ postfix: $ab\times c+$

Reverse direction: given prefix $+\times abc$, rebuild the tree — first symbol is the root; each operator claims the next two complete subtrees: $+(\times(a,b),c)$ — the same tree, recovered.

Prefix / postfix evaluation (stack)

- **Postfix**: read left→right. Push numbers. On an operator, pop two, apply, push the result.
- **Prefix**: read right→left, same rules.
- Order matters for $-$ and $\div$: in **postfix** the *second* value popped is the left operand. In **prefix** (read right→left) the *first* value popped is the left operand.

Worked: stack trace — postfix $5\ 1\ 2\ +\ 4\ \times\ +\ 3\ -$

1. PUSH NUMBERS
$5, 1, 2$ stack: $[5,1,2]$

↓

2. $+$: POP $2,1 \rightarrow 1+2$
push 3 stack: $[5,3]$ *(second pop is the* left *operand)*

↓

3. 4, THEN $\times$: POP $4,3 \rightarrow 3\times4$
push 12 stack: $[5,12]$

↓

4. $+$: $5+12$
push 17 stack: $[17]$

↓

5. 3, THEN $-$: POP $3,17 \rightarrow 17-3$
$=14$ *(the $-$ is where the operand-order trap bites: $17-3$, never $3-17$)*

Euler & Hamilton

- **Euler circuit**: uses every edge exactly once, returns to start. Exists iff connected and all degrees even.
- **Euler path**: every edge once, start $\neq$ end. Exists iff exactly 2 odd-degree vertices.
- **Hamilton circuit**: visits every vertex exactly once. NP-hard in general.

Worked: run the tests on the MST-section graph

1. LIST THE DEGREES
A:3 (B,C,D), B:3, C:4, D:3, E:3

↓

2. EULER CIRCUIT TEST
needs *all* degrees even — four odd vertices $\Rightarrow$ no circuit; Euler *path* needs exactly 2 odd — four $\Rightarrow$ no path either

↓

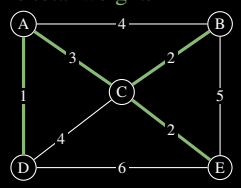
3. HAMILTON — BY INSPECTION
$A \rightarrow D \rightarrow C \rightarrow E \rightarrow B \rightarrow A$ uses edges AD, DC, CE, EB, BA — all exist ✓: Hamilton circuit found *(every vertex once — unused edges are fine. No clean criterion exists, so exhibit one)*

Spanning trees & MST

A **spanning tree** is a tree (no cycles, $n-1$ edges) that touches *all* vertices. A **minimum spanning tree (MST)** is the cheapest such tree by total edge weight. Two greedy builders:

- **Prim's**: start at any vertex. Repeatedly add the cheapest edge joining the tree to an *unvisited* vertex. Grows one connected blob outward.
- **Kruskal's**: sort *all* edges by weight. Add the cheapest edge that doesn't make a cycle; stop at $n-1$ edges.

Both always reach the same total weight.



MST in green $=\{AD, BC, CE, AC\}$, total weight $1+2+2+3=8$.

Worked: Kruskal's trace (global — cheapest edge anywhere)

1. SORT ALL EDGES
$AD(1)$, $BC(2)$, $CE(2)$, $AC(3)$, $AB(4)$, $DC(4)$, $BE(5)$, $DE(6)$

↓

2. TAKE UNLESS IT CLOSES A CYCLE
AD ✓, BC ✓, CE ✓, AC ✓

↓

3. STOP AT $n-1$ EDGES
4 edges for 5 vertices ✓: weight $1+2+2+3=8$ — the run ends here; $AB(4)$ is never even examined

↓

4. WHAT A REJECTION WOULD LOOK LIKE
had the count not closed, next up is $AB(4)$: A and B are already connected (via C) — cycle, skip *(the skip test, shown for when a run does hit it)*

Worked: Prim's trace (local — grow the blob from A)

1. CHEAPEST EDGE LEAVING $\{A\}$
$AD(1)$ blob $\{A, D\}$

↓

2. CHEAPEST LEAVING THE BLOB
options $AC(3), AB(4), DC(4), DE(6)$: take $AC(3)$ blob $\{A, D, C\}$

↓

3. KEEP GROWING — LIST THE FRONTIER EACH TIME
from $\{A, D, C\}$: options $CB(2), CE(2), AB(4), DE(6)$ $\Rightarrow$ take $CB(2)$; from $\{A, D, C, B\}$: options $CE(2), BE(5), DE(6)$ $\Rightarrow$ take $CE(2)$ — all 5 vertices spanned

↓

4. COMPARE
same four edges, weight 8 — different discovery order, same MST: the "both always agree" claim, demonstrated

Legend

Color & notes
functions $f, g,$ sin
variables x, y, n
numbers 0–9
operators $+ - =$
delimiters $( )\,[ ]$
titles in white
labels · results · pitfalls
$\boxed{u}$ / $\boxed{du}$ substitution

Greek
constants (fixed values):
$\pi \approx 3.14159$
γ fluid wt density
ρ weight per length
variable & operators:
θ angle (varies)
Δ "change in"
Σ sum of terms
notation, not values:
$\Theta\,\Omega\,\omega$ growth rate, like O

Calculus & relations
$\int$ integral
$\sum$ sum
lim limit
∞ infinity
$\to$ approaches
$\Rightarrow$ implies
$\Leftrightarrow$ iff
$\sqrt{\ }$ root
$\approx$ approx
$\neq \leq \geq$ compare
$\pm \mp$ plus / minus
$\cdot \times \div$ mult / div
$\circ$ compose

Vectors
$\langle\,\rangle$ components
$\|\vec{v}\|$ magnitude
$\vec{v}$, $\hat{v}$ vector, unit
$\mathbf{i}\,\mathbf{j}\,\mathbf{k}$ basis
$\vec{u}\cdot\vec{v}$ dot product
proj projection

Logic & sets
$\forall$ for all
$\exists$ exists
$\neg$ not
$\wedge \vee$ and / or
$\oplus$ xor / sym. diff
$\in \notin$ in / not in
$\subseteq \subset$ subset
$\cup \cap$ union / inter.
$\setminus$ set minus
$\equiv$ equiv / congruent
$\mid$ divides / $|x|$ / s.t.
$\lceil\,\rceil$ ceiling
$\binom{n}{k}$ choose
deg degree
mod modulo